QoL-related data clustering and classification for predicting QoL deterioration indicators for immunotherapy cancer patients

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Abstract

Cancer patients undergoing immunotherapy experience diverse and evolving Quality of Life (QoL) trajectories that significantly impact treatment outcomes. This study presents a comprehensive machine learning framework for predicting QoL deterioration by integrating patient clustering with symptom classification over an 18-month follow-up period. We analyzed longitudinal data from the QUALITOP real-world cohort, comprising cancer patients from two hospital partners in Spain and Portugal. Our methodology introduces several key innovations: (1) a temporal aggregation approach that consolidates variable-length patient trajectories into meaningful predictive features while addressing missing follow-ups; (2) direct analysis of patient-reported FACT-G item responses rather than composite scores, enhancing model explainability while achieving 89.5% predictive accuracy; (3) dual-perspective analysis that simultaneously clusters patients into distinct QoL trajectories and identifies symptom patterns, addressing a gap in prior research that examined these dimensions separately; and (4) robust validation through ensemble comparison of seven clustering algorithms and six classification methods. Unlike studies relying on controlled clinical trial data, our approach leverages real-world evidence to reflect actual clinical settings and their inherent complexities. The resulting prediction model provides caregivers with interpretable aggregated indicators for early identification of patients at risk of QoL deterioration, enabling timely intervention during immunotherapy treatment. Our findings demonstrate that patient-centric, longitudinal analysis of QoL data can effectively support clinical decision-making in oncology, with implications for personalized cancer care and improved patient outcomes.

We are interested in this work, into clustering and classifying cancer patients' symptoms under immunotherapy treatment, in order to predict future steps and medical events within a proposed medical project platform. One key health dimension is on focus: patient Quality of Life. We present some existing clustering and correlation methods related to this dimension. Within the Qualitop project, we analyzed collected datasets under agreement mainly from 2 hospital partners (Spain, and Portugal). Our analysis focuses on QoL-patient data and some key clinical data. We aggregate disease evolution attributes through time (18 months). We use supervised and unsupervised learning algorithms, and trigger conclusions based on calculated aggregated features of QoL. A prediction model is proposed based on training data. The model helps caregivers interpret patient symptoms based on aggregated indicators.

1 Introduction

Immunotherapies have improved in a significant way the clinical results of various cancers (Liu et al., 2022). Immune Checkpoint Inhibitors (ICI) and Chimeric Antigen Receptor T-cell (CAR-T-cells) are the two main axes of immunotherapies (Hosseinkhani et al., 2020). These immunotherapies cause Immune-related Adverse Events (Ir-AE) to patients (Pozzessere et al., 2024). Quality of Life reporting is important in decision-making for treatment options. It mainly helps the care providers in identifying and prioritizing the most important problems and hidden problems of patients (Gour and Chaudhary, 2023). Adverse Events and their interaction with health-related Quality of Life (HRQoL) are important to predict for better prevention of the Adverse events and for taking more effective decisions by caregivers (Lustberg et al., 2023; Belzer et al., 2024; Wenzel et al., 2021).

There are several clinical trials and real-world datasets related to immunotherapies, but they do not properly reflect the real-world settings (Vinke et al., 2023). The Qualitop project proposes a real-world cohort (accessed under agreement) consisting of collected data from several European hos-

pitals over 18 months. Each collected time data consists of repeated HRQoL data evaluations by patients and professionals. This time-related data collection helps in the study of patients' time trajectories and examines the causes during the time of HRQoL changes (Cuzzocrea et al., 2023).

We aim in this work, to predict the causes of future Adverse Events (AE) and QoL deterioration from collected cross-sectional and longitudinal QoL-related data. It was reported in the literature that there is a significant discordance between clinician and patient-reported data (Chen et al., 2022). For this, we propose to use in this work, mainly the QoL patient data, and some key clinical data to try to understand the causal mechanisms of QoL deterioration, considering the patient view. Our data focus on patient data and a few clinical data related mainly to medications and therapies.

The main goals of this work are to

- Characterize clusters of cancer patients using a real-world cross-cohort dataset
- Identify determinants (or predictive indicators) of HRQoL and their correlations
- analyze the groups of patients and the QoL found key attributes
- Propose a prediction model based on the findings
- Help into consolidating literature QoL determinants of cancer disease using a method of temporal aggregation of raw QoL patients' data

This paper is organized as follow we define first, the standards of assessing QoL. Second, we present the state-of-the art of AI and statistical methods to predict the QoL determinants for Cancer patients. In addition, we present the correlation approaches of QoL-related factors. Third, we present our method for temporal aggregation of patients longitudinal data. Also, we define the used clustering and classification models of QoL-related symptoms. Finally, we present our experimental findings about the identified clusters of patients and QoL-related symptoms correlations.

2 QoL Assessment

QoL is a set of subjective perceptions that are hardly defined and measured in a objective way. Different definitions are provided in the literature (Gour and Chaudhary, 2023). We refer in our study to the definition of The World Health Organization (WHO) for QoL as *the perception of an individual of their position in life*. Determinants of QoL are related to the context of culture, the individuals' goals, expectations, standards, and concerns (Haraldstad et al., 2019)

There are mainly four dimensions of HRQoL components which are: *(i) Physical health* related mainly to disease symptoms and side effects, *(ii) Mental health* related to a spectrum from well-being, psychological distress to psychiatric diseases, *(iii) Social health* related to quantity and quality of social interactions and *(iv) Functional health* including physical activity and self-care (Gour and Chaudhary, 2023).

QoL-related symptoms have been standardized and assessed by questionnaires and their potential summarized measures in the literature as several categories such as :

- Generic HRQoL (Bottomley et al., 2021) which are generic instruments for any disease or any population that can be useful for comparison among disease or populations
- Global HRQoL score (Bottomley et al., 2021) which describes a composite score of a patient overall HRQoL based on several dimensions (*e.g.* physical, mental, emotional, social), they can be disease-specific or generic instruments
- Profile measures (Kaplan and Hays, 2022) is used for better granular analysis, it calculates individual scores for each dimension of the HRQoL (*e.g.* SF-36) instead of using a global HRQoL score,
- disease-targeted measures (Kaplan and Hays, 2022) which captures symptoms and impacts related to a specific disease (*e.g.* FACT-B for breast cancer)
- Quality-adjusted life year (QALY) (Kaplan and Hays, 2022) (*e.g.* EORTC QLQ-C30) which combines life expectancy with QoL in order to assess the cost-effectiveness relation of medical interventions
- Individualized measure of QoL (IQOL) (Pai et al., 2020) (*e.g.* SEIQoL) which allows patients to rate the most important dimensions of their life for a personalized QoL profile

- Standard patient-reported outcome measures (Kuspinar et al., 2020) (*e.g.* PROMIS) which consists on health and QoL questionnaires completed directly by patients without interpretation of clinicians

We present an overview of the standardized QoL instruments which are questionnaires and calculated scores used for assessing QoL-related symptoms. Most of the studies have used EORTC and secondly FACT questionnaires (see Table 6). Table 6 lists some questionnaires assessing QoL where Q: Questionnaire and AD: Assessment Dimension. It is to note that EORTC QLQ and FACT tools refer to many assessment tools versions for different aspects and cancer types (*e.g.,* EORTC QLQ-C3, EORTC-QLQ-LC43, EORTC-QLQ-BR23, FACT-Bl, FACT-G, FACT-Lym, FACT-M).

(Vinke et al., 2023) presents the assessment domains of our QUALITOP project questionnaire including FACT-G which is a profile and disease-targeted QoL assessment tool.

We report in the following section, the QoL-related studies consisting mainly of the clustering of cancer patients and their QoL-related symptoms. Also, we present correlation methods of the QoL-related symptoms.

3 Clustering of QoL-related symptoms

Keywords of search (2021-14/01/2025)

"clustering QoL immunotherapy", ""clustering algorithm" "QoL" "immunotherapy"", "clustering quality of life immunotherapy"

In this section, we present studies that proposed to cluster cancer patients according to their QoL using adapted QoL-related questionnaires and scales. This is fundamental to grasp in order to understand the actual state of selecting determinants/indicators of QoL in actual studies.

Breast cancer symptoms are often described by patients as more than one symptom at a time known as cluster symptoms. For this (Al Qadire et al., 2021), studied the *symptom clusters* for breast cancer patients receiving a combination of therapies (chemotherapy, radiotherapy, hormonal, and *immunotherapy*). The goal is to allow the management of symptoms as clusters to reduce the one-by-one symptom-related side effects and medication costs. Symptom clusters are also used to predict the mean score of QoL. Exploratory Factor Analysis (EFA), and multiple linear regression are used for these purposes. 29 symptoms are included in the Factor Analysis. Five clusters are identified: *(i)* psychological (*anxiety, feeling tense, depressed mood, worrying, irritability, despairing about future, and nervousness*), *(ii)* treatment side-effects (*burning/sore eyes, dry mouth, loss of hair, shortness of breath, sore mouth, and acid indigestion*), *(iii)* nausea and vomiting (*nausea, vomiting, lack of appetite, diarrhea, and dizziness*), *(iv)* pain (*lower back pain, headache, abdominal pain, constipation, shivering, sore muscles, tingling hand or feet, and difficulty in concentrating*), and *(v)* fatigue (*tiredness, lack of energy, sore muscles, and difficulty sleeping*). The identified clusters together explained about 53.4% of the total variance in symptoms, the higher the variance is the more the clusters are more meaningful and distinct. The mean score of QoL among patients was 20.9 out of 30 (maximum score). Linear regression is used to identify which cluster would predict the patients' QoL. Psychological, nausea/vomiting and pain clusters were the three high predictors of the total score of QoL (Al Qadire et al., 2021).

In (So et al., 2021), authors propose a systematic review between 2005 and 2020 of existing symptom cluster studies for breast cancer patients before, during, and after cancer treatments (mainly chemotherapy, radiotherapy, hormonal therapy, or surgery). It outlines the lack of systematic reviews and meta-analyses summarizing the symptom cluster findings of different studies. Also, it discusses the symptom cluster stability at different time points. Out of conclusions, 3 symptom clusters *before treatment* are common in 4 out of 11 studies. The clusters contain *(i) Pain/Fatigue/Sleep disturbance*, *(ii) Menopausal symptoms*, and *(iii) Psychological symptoms (sadness, worry, anxiety, depression)*. Before cancer treatment, at least two symptoms of the *psychological cluster* co-occur in patients within 8 studies. Four studies out of the 8 show an association between symptoms of the first cluster and the third cluster. The most reported clusters for *undergoing treatment symptoms* are *(i) Gastrointestinal (nausea, lack of appetite)*, *(ii) Pain/Fatigue/Sleep disturbance*, and *(iii) Psychological (anxiety, depression, worry, sadness, nervousness, and irritability)* symptoms. Four cluster symptoms are reported for *post treatment*: *(i)* Fatigue/Sleep disturbance, *(ii)* Psychological (depression, anxiety), *(iii)* Gastrointestinal (nausea, lack of appetite, diarrhea), and *(iv)* Menopausal symptoms. Additional correlated symptoms within

Table 1: QoL and related symptoms Assessment tools for Cancer patients

Q	AD
EORTC QLQ tools (Machingura et al., 2022; Jung and Yun, 2022; Huang et al., 2019; Bjoern et al., 2019; Boekhout et al., 2021; McDowell et al., 2021; de Rooij et al., 2021; Kim et al., 2021; Li et al., 2022; Durán-Gómez et al., 2023; Zhang et al., 2023; Savander et al., 2023; Yang et al., 2023; Bland et al., 2021)	QoL
FACT tools (Mohamed et al., 2021; Cuzzocrea et al., 2023; Vinke et al., 2023; Durán-Gómez et al., 2023)	QoL
FACT-Cog (Durán-Gómez et al., 2023)	Cognitive complaints
SF-12 QOL & FACT-C14 (Parc et al., 2019)	QoL
FAACT (Bland et al., 2021)	Anorexia/Cachexia
PROMIS (McDowell et al., 2021; Lee, 2022)	Self-reported QoL
HADS (Mohamed et al., 2021; Boekhout et al., 2021)	Anxiety & depression
HSQ (Jung and Yun, 2022)	Health status
SAT (Jung and Yun, 2022)	Self-management strategy
BCNAS-32 (Mohamed et al., 2021)	Unmet needs
GSE (Mohamed et al., 2021)	Optimistic self-control beliefs
UW-QOL (Rogers et al., 2021; Mohamed et al., 2021)	QoL
LARS (Kim et al., 2021)	QoL
RSCL (Al Qadire et al., 2021)	Symptoms including QoL
QLI-C (Al Qadire et al., 2021)	QoL
MFI (Boekhout et al., 2021), FSI (Durán-Gómez et al., 2023)	Fatigue
SCQ (Boekhout et al., 2021)	Comorbidities
BDI (Durán-Gómez et al., 2023), SDS (Li et al., 2022)	Depression
STAI (Durán-Gómez et al., 2023)	Anxiety
PSQI, ISI (Durán-Gómez et al., 2023)	Insomnia

these symptom clusters could vary depending on the study parameters.

(de Rooij et al., 2021) proposes a study for analyzing the correlation between symptoms of 7 cancer types (colorectal cancer, breast cancer, ovarian cancer, thyroid cancer, chronic lymphocytic leukemia, Hodgkin lymphoma, and non-Hodgkin lymphoma). The study concerns survivors of cancer who received different cancer treatments (radiotherapy and chemotherapy). It used 9 symptom scales from the EORTC QLQ-C30 questionnaire (ie, *fatigue, pain, dyspnea, constipation, lack of appetite, diarrhea, nausea/vomiting, cognitive symptoms*, and *emotional symptoms*). It performs a *network analysis* and partial correlations in the networks to cluster symptoms. Each network corresponds to a cancer-type dataset sample. It makes conclusions about *(i)* correlations of symptoms, *(ii)* the difference in symptom correlation from one type of cancer to another, *(iii)* symptom correlation according to the cancer treatment, and *(iv)* short-term and long-term symptom comparison. Out of the several conclusions, the study shows the presence of a prevalent cluster (*fatigue, pain, emotional symptoms, appetite loss, and dyspnea*) for all cancer types and treatments for short-term and long-term survivors. Also, the well-known cluster (*fatigue, insomnia, and depression*) in the literature was confirmed by this study.

(Boekhout et al., 2021) proposes a multi-center cohort study in order to evaluate the HRQoL and other symptoms. The cohort concerns long-term advanced *melanoma* survivors treated with an ICI *immunotherapy* treatment (*i.e.,* ipilimumab) without the addition of other systemic (affecting cells all over the body) therapies. Spearman's correlation, Mann-Whitney or Kruskal-Wallis statistical tests are used to compare QoL-related questionnaire scores with other symptom scores. Associations between QoL-related scores and immunotherapy cumulative dose, and demographic characteristics are studied. A comparison is also performed between *(i)* survivors and matched controls without cancer, *(ii)* survivors after and before 36 months of follow-up, *(iii)* with and without brain metastases, *(vi)* with more than 2 comorbidities and without. The *36 months* threshold corresponds to the risk of recurrence and the time a patient is considered to be cured. Analyzed domains include nine symptomatic domains (fatigue, nausea/vomiting, pain, dyspnea, insomnia, appetite loss, constipation, diarrhea, and financial impact), and one domain of global health and quality of life (QoL). From conclusions, melanoma survivors with immunotherapy

had higher symptom burden than matched controls in terms of: *(i) fatigue*, *(ii) dyspnea*, *(iii) diarrhea*, and *(iv) financial impact*. No significant difference on global QoL is reported between the two groups, mentioning a response shift.

(McDowell et al., 2021) clusters the Human Papillomavirus-associated Oropharyngeal Cancer survivors patients treated with chemo-radiotherapy. Survivors are patients who are disease-free after 12 months of radiotherapy or more. It uses auto-clustering statistics (log-likelihood measure, Bayesian Information Criterion, silhouette measure) to identify functioning-based subgroups. It identifies two main clusters with an acceptable quality (average silhouette of 0.6): *(i)* high-functioning subgroup (69%) which consists of patients with better functioning on all QoL-related and symptom burden scales and the rest of patients as *(ii)* low-functioning subgroup (31%). Out of conclusions, the study reports a significant difference between clinicians and patients-reported severe toxicity (Adverse Reactions). In comparison, clinicians reported that 14% of patients have at least one severe toxicity, and almost 50% of patients reported at least one severe toxicity. No significant variation between the two clusters of patients was determined between patients' sociodemographic (*i.e.,* age, gender, education, employment, relationship status), tumor (stage), and treatment factors (*e.g.,* chemotherapy or unilateral radiation). Large mean differences are reported between the two clusters in terms of *global QoL*, and some symptom scales (*fatigue, pain, dyspnoea, insomnia, and appetite loss*).

(Mohamed et al., 2021) tries to identify the deficiencies in healthcare services and support of vulnerable bladder cancer patients (called with unmet needs). Hierarchical agglomerative (HA) and partitioning (PC) clustering are performed to cluster patients in two groups *high* and *low/moderate unmet needs*. Patients fill out surveys of unmet needs, QoL (FACT-Bl), and other dimensions. Out of conclusions, the two clustering methods show no significant difference in patients' demographic and clinical characteristics between the two groups. However, with HA clustering, a *high-unmet needs* cluster correlates with worse QoL, emotional adjustment, and lower personal and social resources than with PC clustering. Also, with the HA clustering, the *high-unmet needs* group has significantly higher needs as *psychological, health system, daily living, sexuality, logistics, and communication* than with PC. The authors recommend validating the stability of the identified clusters by examining them longitudinally in future research.

(Li et al., 2022) evaluates the association between depression disorder and QoL for patients after combined immuno-chemotherapy treatment. The patients have oncogene-driver negative non-small cell cancer (NSCLC). Descriptive analysis (SPSS) and Cox regression are used to evaluate the correlation. Two groups are identified: *with and without depression*. Out of conclusions, there is no significant difference between the two groups in terms of demographics (*e.g.,* age, sex, education) and clinical characteristics (*e.g.,* pathological type, tumor stage). There is a negative association between depression disorder and QoL for NSCLC patients following a combined treatment of chemotherapy-immunotherapy. The conclusions made in this study are based on a small sample of patients (67 patients).

(Machingura et al., 2022) proposes a clustering study that confirms previous literature findings of Martinelli et al.. The findings infirm that the 15 scales of EORTC QLQ-C30 QoL questionnaire are interrelated and are grouped in three main clusters: *(i) physical* (role-functioning, physical-functioning, social-functioning, fatigue, pain, and global-health status, *(ii) psychological* (emotional-functioning, cognitive-functioning, and insomnia), and *(iii) gastrointestinal* (nausea/vomiting and appetite loss). The tested dataset includes patients with 17 cancer types and different cancer therapies (systemic or not) at baseline. Spearman-rank correlations are used to make associations between QoL scales. The agglomerative hierarchical analysis is used for clustering. Ward's method is used for clusters' similarity. Out of conclusions, the clusters were consistent considering different characteristics (*i.e.,* different cancer types, sociodemographic and clinical characteristics of cancer patients). The global health status(GHS) is strongly related to the physical-functioning cluster. Patients with *melanoma* have high scores on functioning scales and low ones on symptom scales. A strong correlation is observed between *fatigue* and role-functioning/physical-functioning scales.

(Yin et al., 2022) investigates the effects of *fear of cancer recurrence (FCR)* on *immunotherapy* efficacy and *QoL* for non-small cell lung cancer (NSCLC) patients during the COVID-19 pandemic in China. Data is collected at two-time points. It clusters the patients into two groups: *high-FCR*

and low-FCR using the CWS questionnaire score. Statistical analyses (t-test, X^2, rank-sum test) are used to find the differences between the two clusters. Linear regression is used to study the correlations. Out of conclusions, there is no significant difference in demographic and medical characteristics (*e.g., age, sex, education, cancer stage, immunotherapy drugs, therapies*) between the two groups. It affirms that the *high-FCR group* has worse immunotherapy efficacy than the *low-FCR*. QoL scores decreased for both groups at the study endpoint compared to baseline scores. For the high-FCR group, QoL is significantly lower during immunotherapy than the low-FCR group. There is no significant difference in Adverse Events for the two groups. The authors conclude the need to consider and alleviate the level of FCR in NSCLC patients. However, they mention that the conclusions should be further validated by more studies. In fact, the analyzed data is from a single center, a small sample, and contains only one follow-up after the baseline.

(Savander et al., 2023) proposes to use different methods of dimensionality reduction and clustering for radiotherapy cancer patient groups analysis. It includes as a clustering feature the QoL using the mean of QLQ-C30 questionnaire summary score. It captures electronic patient-reported outcomes(called ePRO) via a smart healthcare application (called Haiku Health). The captured data are both categorical and continuous data. It uses two methods of dimensionality reduction: *(i)* factorial analysis of mixed data (called FAMD) and *(ii)* uniform manifold approximation and projection (called UMAP). It employs two clustering algorithms: *(i)* k-means and *(ii)* hierarchical clustering. It combines the dimensionality reduction and clustering methods into 3 main clustering methods. It uses a combined *internal indices* measure for evaluating the goodness of the clusters. For cluster interpretation, it uses statistical summarization and decision trees. Combining UMAP and k-means gives the best results according to the combined internal indices measure. According to their cluster interpretation (with decision trees), the most important feature for clustering (with UMAP & k-means) is QLQ-C30 summary score.

(Zhang et al., 2023) proposes to cluster advanced Colorectal Cancer (CRC) patients using the trajectories of their QoL (longitudinal QoL data at 7 time points). It used an unsupervised learning algorithm which is growth mixture modeling (GMM) to detect unobserved patient clusters. The identified clusters are evaluated considering clinical-epidemiological characteristics and overall survival (OS). Authors conclude that GMM algorithm can evaluate the longitudinal QoL characteristics of individuals effectively.

(Cuzzocrea et al., 2023) proposes as part of our Qualitop project, two techniques adapted to big data analytics to discover hidden temporal patterns in QoL attributes over cancer patients. It studies lung cancer patients treated with immunotherapy including multi-therapies. It constructs QoL tables and their relative frequency table streams over four time units (3, 6, 12, and 18 months). Further, it performs clustering over each frequency table using the k-means algorithm. It also performs a similarity function (Cosine function) over different clustering (groups of clusters corresponding each to a frequency table). In the experiments, it considers one dimension of QoL which is physical well-being. It makes mainly singular conclusions from statistics of the clusterings about Immune-Related Adverse Events (irAEs), monotherapy/multitherapy impacts on QoL, and the evolution of QoL attributes over time.

(Yang et al., 2023) proposes to cluster symptoms in order to identify central symptom clusters and study their effect on QoL. The study concerns lung cancer patients receiving *immunotherapy* treatment. Principal Component Analysis (PCA), Spearman correlation analysis, and multiple linear regression are mainly used to identify and describe symptom clusters and to study the influence of these clusters on QoL. Five symptom clusters are identified: *(i) emotion-related*, *(ii) lung cancer-specific*, *(iii) perceptual, (iv) skin, and (v) neural*. The five clusters explained about 55.81% of the global symptom variance. Using the covariate adjustment technique revealed that the identified 5 symptom clusters are negatively correlated to QoL. Also, the three clusters (lung cancer-specific, emotion-related, and perceptual symptoms) are negative predictors of QoL. Linear regression model reveals that skin symptom cluster and global QoL have no significant correlation. The other clusters have significant correlations with the scores of QoL dimensions. There is no significant difference in QoL for patients who received more than 3 immunotherapy treatments with those who received less.

(Durán-Gómez et al., 2023) identifies subgroups of psycho-neurological symptoms (PNS) and their associations with clinical variables. Also, it studies the effect of the PNS symptoms on QoL. The study

Table 2: Clustering and correlation methods for QoL-related symptoms

Study	Methods
(Al Qadire et al., 2021)	Exploratory factor analysis, Multiple linear regression
(de Rooij et al., 2021)	Regularized partial correlation network analyses, polychoric correlations
(Boekhout et al., 2021)	Mann-Whitney or Kruskal-Wallis tests, Spearman's correlation
(McDowell et al., 2021)	Two-step cluster analysis, log-likelihood distance measure
(Bland et al., 2021)	Descriptive analysis
(Mohamed et al., 2021)	Hierarchical agglomerative, partitioning clustering, Ward's agglomeration method, Elbow method
(Li et al., 2022)	Descriptive analysis, Cox regression analysis
(Lee, 2022)	k-means, Elbow method
(Machingura et al., 2022)	Agglomerative hierarchical, Spearman-rank correlations, Ward's method
(Jung and Yun, 2022)	k-means, PCA
(Savander et al., 2023)	k-means, Hierarchical clustering, statistical summarization and decision trees
(Zhang et al., 2023)	Unsupervised growth mixture modeling (GMM), Cox regression, Concordance indices
(Cuzzocrea et al., 2023)	k-means, Cosine similarity measure
(Yang et al., 2023)	Exploratory Factor Analysis, Multiple linear regression, Spearman correlation
(Durán-Gómez et al., 2023)	Factorial Principal components analysis, Pearson coefficient

concerns a sample of breast cancer women survivors with different treatments (mainly chemotherapy combined with other therapies including immunotherapy). The data is collected at a single time point post-treatments. Principal component analysis is used to cluster symptoms. Pearson r coefficient is used for correlations of factors. Three PNS clusters are identified with a 73% of variance in symptoms: *(i) two aspects of anxiety*, *(ii) cognitive impairment, pain and fatigue*, and *(iii) sleep disorders*. Two QoL clusters are identified with a 64.9% of variance: *(i) Functional-physical* and *(ii) cognitive-emotional*. A relationship exists between sleep disorders cluster, its negative impact on QoL and chemotherapy treatment. The most prominent symptoms on patients are *fatigue, insomnia, anxiety, cognitive impairment and depressive symptoms*.

Table 5 summarizes the clustering and correlations methods used to study the interrelationships between QoL scales and related symptoms.

Table 7 lists some QoL-related datasets in the literature, where S: Study, NP: Number of Patients, CaT: Cancer Type, T: Treatments, DP: Dataset Provenance, M: Follow-up in months. Dataset provenance can be from questionnaire, an existing dataset or clinical trials, or from self-reported outcomes. Period in months is not specified ('-') if the number of follow-up is not mentioned clearly in terms of months, or if there is no follow-up.

Please refer to Table 6 for the QoL assessment dimension of the literature works.

Most of these works used scores of HRQoL, which is informative for statistical models but it may lack explainability for machine learning models. Also, clustering have been performed in most works related to a fixed symptom or group of symptoms (*e.g.* depression, psycho-neurological symptoms (PNS)), which lacks general information on the global clusters of patients (independently of the fixed symptom group). Finally, the literature works focused either on grouping patients or grouping symptoms' patients. It is important to consolidate the findings of both patients groups and symptoms groups.

We present in the following section our prediction approach about QoL-related symptoms using Qualitop datasets.

4 Prediction scenario through QUALITOP datasets

We present our approach of a prediction model based on clustering and classification of temporal patient attributes aggregation. First, data is collected of several time questionnaires and hospitals. Second, data is coded and cleaned. Then, all time questionnaires are mapped to represent a one questionnaire for all times. Key attributes are aggregated for all time questionnaires for each patient. Supervised learning models are applied to the dataset. Unsupervised learning models are also used. A mixing of all hospital datasets is highly considered in our architecture, but it is not compulsory.

First, we present a descriptive analysis of 2 selected Qualitop datasets, that we have access to with agreement. Secondly, we present our prediction scenario and identify clusters through these datasets.

Table 3: State-of-the art QoL-related datasets

S	NP	CaT	T	DP	M
(Machingura et al., 2022)	20,066	17 types	-	Multi-source clinical trials	-
(Jung and Yun, 2022)	254	-	-	Multi-hospital questionnaire (South Korea)	6
(Zhang et al., 2023)	750	Colorectal	Chemotherapy	Clinical trials (Canada & Australia)	4
(Mohamed et al., 2021)	159	Bladder	-	Bladder Cancer Advocacy Network questionnaire	-
(Bjoern et al., 2019)	85	Rectal	Surgery	Hospital questionnaire (Denmark)	8
(Kim et al., 2021)	283	Colorectal	Surgery	Hospital questionnaire (Korea)	32
(Savander et al., 2023)	281	Cancer	Radiotherapy	7 hospitals self-reported outcomes	-
(Lee, 2022)	11	Advanced	-	Kaiku Health self-reported outcomes	3
(Yang et al., 2023)	240	Lung	Immunotherapy	Multi-center questionnaire (China)	-
(Al Qadire et al., 2021)	516	Breast	Multi-therapy	3 hospitals questionnaire (Jordan)	-
(Huang et al., 2019)	23	Metastatic /advanced	-	multi-center clinical trials	8
(Boekhout et al., 2021)	89 and 265	Advanced melanoma survivors	Immunotherapy (ipilimumab)	Multi-center questionnaire (Netherlands & Belgium)	39
(Durán-Gómez et al., 2023)	239	Breast	Multi-therapy	Hospital questionnaire (Spain)	-
(Li et al., 2022)	33	Non-small cell Lung cancer	Combined immuno-chemotherapy	Hospital questionnaire (China)	-
(Kang et al., 2021)	291	Cervical Carcinoma	Immunotherapy	Samples of TCGA dataset	-
(Feng et al., 2022)	298	Urothelial	immunotherapy	IMvigor210 dataset	-
(Chen et al., 2022)	2456	Gastric	-	12 cohorts	-
(McDowell et al., 2021)	136	12-month HPV-OPC survivors	Chemo-radiotherapy	Hospital questionnaire (Australia)	6
(Bland et al., 2021)	162	Cancer	-	Hospital questionnaire (Australia)	-
(de Rooij et al., 2021)	1330	Survivors of 4 cancer types	Surgery, chemotherapy, radiotherapy	Cross-sectional survey data	-

4.1 Descriptive Analysis of QUALITOP datasets

4.1.1 IPOL Portugal dataset

Portugal dataset is provided by IPOL (Instituto Portugues de Oncologia Lisboa) Institute. All patients in the dataset suffer from Lymphoma cancer and received immunotherapy treatments (ICIs, CAR-T-cells, and other immunotherapies). The data dictionary contains 425 attributes. The dataset consists of 49 patients and 483 attributes. The dataset (excel file) is in the form of matrices of data, each matrix is related to a Qualitop domain (*i.e.,* questionnaire or *'Form'*). For IPOL, there are 16 domains for both datasets (Vinke et al., 2023). QoL-related indicators are mainly in the 8th and 9th Forms corresponding respectively to baseline and several follow-up questionnaires. The follow-up questionnaire is repeated from 3 months to 24 months, every 3 months. A progression questionnaire is also completed providing final values after 24 months (*e.g.,* death status of the patient). It is to be noted that IPOL data tend to be very few (about 4 patients responding to the QoL questionnaire) after 18 months.

4.1.2 HCB Spain dataset

Spain dataset is provided by HCB(Hospital Clinic de Barcelona) hospital. All patients in the dataset suffer from Melanoma and have received ICI immunotherapy treatment. There are two types of data for analysis: retrospective and prospective

(Vinke et al., 2023). QoL-related data exists only within prospective data. Within the last, There are 55 patients and 459 attributes. The HCB dataset contains 14 domains (Vinke et al., 2023).

IPOL Immunotherapy and HCB datasets are compared in the Figure 1 in terms of their respective attributes statistics. We specify *(i)* the total number of attributes in dataset, *(ii)* the number of attributes for baseline questionnaire, *(iii)* the number of 60% answered attributes for baseline, *(iv)* the number of follow-up attributes, *(v)* the number of 60% answered attributes for follow-up. According to Figure 1, there is a 9 or 10 difference in baseline and follow-up attributes for the two datasets. Questionnaire attributes in HCB and IPOL are filled at least more than 60%.

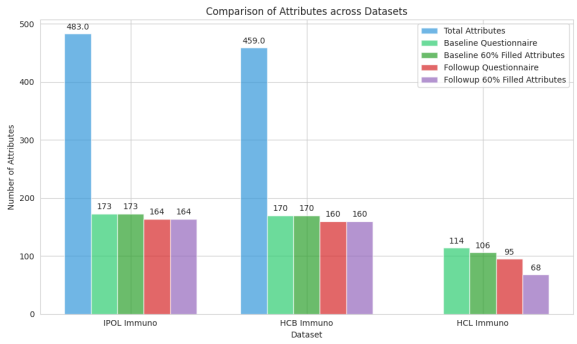

Figure 1: Comparison of Attributes across 3 Qualitop datasets

We present in Figure 2 statistics about IPOL immunotherapy and HCB datasets in terms of patients. We specify as follow: *(i)* the number of patients, *(ii)* the number of 60% filled responses (patients) for baseline, *(iii)* the number of patients for each follow-up (Months 3, 6, 12, 18, 24) and *(iv)* the number of 60% filled responses for each follow-up (Months 3, 6, 12, 18, 24).

Figure 2 shows the homogeneity of datasets through time (baseline and follow-ups). IPOL dataset lacks the filling of the questionnaire properly. For example, 49 patients are supposed to have filled the questionnaire, but only 15 of them have filled 60% of the QoL-related questionnaire at baseline and 14 of them have filled it after 3 months. Two patients have filled 60% of the QoL-related questionnaire after 18 months. HCB dataset is more homogeneous through time.

Due to its homogeneity through time, we choose to use HCB dataset as a starting point for our Qol-related attribute selection.

A mapping of attribute labels between IPOL and

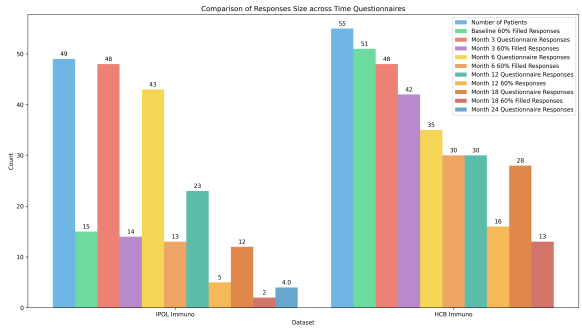

Figure 2: Comparison of Patients statistics across 2 Qualitop datasets

HCB datasets is applied. A partitioning of data according to time questionnaire is performed allowing an analysis for each time questionnaire separately, if required. Before analysis, it is important to grasp the dimensions of datasets, and further to choose the dimensions that should be aggregated or analyzed separately. For this, we present in the following the Qualitop datasets dimensions.

4.2 Qualitop datasets dimensions

Qualitop cohort can be partitioned/aggregated according to different domains which are: *(i)* Forms (Questionnaires), *(ii)* therapy types (ICIs, CAR-T-cells), *(iii)* hospital of inclusion (HCB, IPOL), *(iv)* time questionnaire (baseline, 3-months follow-up, *etc.*)

We consider all therapy types and all-time questionnaires. We consider time (called 'event name' in Figure 3) as a *dimension* for aggregation. It is important to average all time questionnaires since Cancer-related QoL for patients is considerably variable from one-time point to another due to the prompt and delusion effects of the disease and its related therapies.

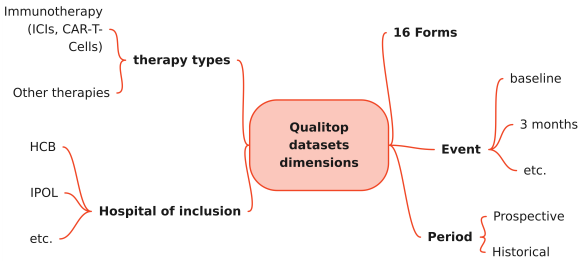

Figure 3: Qualitop datasets dimensions

4.3 Data aggregation

Qualitop datasets are represented as matrices. Each matrix is related to a questionnaire domain (Vinke

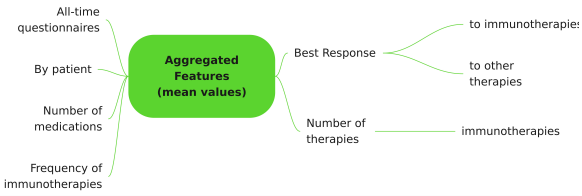

Figure 4: Aggregated Features

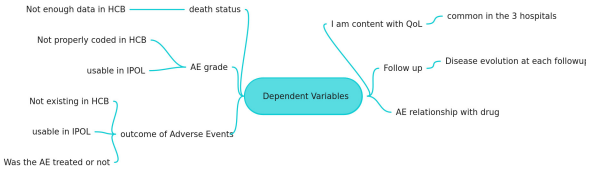

Figure 5: Dependent Variables

et al., 2023) (called Forms). We process these matrices of data in order to have one line per patient. We aggregated multiple values of attributes per patient for all time questionnaires. More specifically, we mapped baseline attributes with follow-up attributes in order to perform aggregation.

Data aggregation refers to the process of gathering, reformatting, and presenting *raw data* in a summarized form for data sharing and analysis. Raw data can be aggregated generally by time, by location, or by data source. Aggregated data have been used for describing new and unusual patterns (Wen, 2020).

We present in the following the features that are aggregated in our preprocessing.

4.3.1 Aggregated features

Six features are aggregated from four raw attributes (see Figure 4) which refer to mean values for each patient for the whole survey: *(i)* disease status after all immunotherapies, *(ii)* disease status after other therapies (chemotherapy, radiotherapy), *(iii)* number of medications, *(iv)* number of immunotherapies, *(v)* number of other therapies, and *(vi)* frequency of immunotherapies.

4.3.2 Aggregated dependent variables

As for aggregated features, mean values are calculated for each patient for HCB and IPOL combining all time questionnaires.

There are six potential variables to predict (see Figure 5). One of them is assessed by patients which is: *(i)* degree of QoL patient contentment, and the rest is assessed by professionals: *(ii)* disease status or response to treatment (*1, Stable disease* | *2, Progressive disease* | *3, Partial response* | *4, Complete response* | *5, Not evaluated*), *(iii)* if the Adverse Events is in relationship with administrated drugs or not, *(iv)* outcome of Adverse Events (AE): if the AE are treated successfully or not, *(v)* highest grade of Adverse Events, and *(vi)* death status.

Three out of the six attributes are selected according to their missing or ambiguous values for the two collected datasets, as described in Figure 5. The three aggregated dependent variables are QoL contentment (I am content with QoL), disease status (follow-up), and AE potential relationship with drugs.

4.4 Data selection

As we mentioned before, HCB dataset is considered as a starting point for data cleaning. The raw common attributes are the id of patient in the study (anonymous id), gender of the patient, date of birth, cancer type and its weight. We do not include the height due to few data for HCB dataset for this attribute. Also, heart rate and arterial pressure data are fewer than QoL-related data, which would imply reducing the number of patients. We exclude them also in our study. All QoL raw questionnaire attributes are included as a starting point (excluding raw attributes converted to aggregated ones). All aggregated features and dependent variables are included. Allergies, blood analysis, and other clinical data are excluded in order to focus on the impact of patient assessed QoL data on the dependent variables.

For missing values, we choose to delete attributes rather than patients. We perform the cleaning of QoL-related attributes with more than 60% threshold of missing values. The IPOL dataset after preprocessing consists of 58 time patient data (corresponding to 16 patients). The HCB dataset consists of 152 time patient data (corresponding to 52 patients). The two datasets contain 53 attributes (see annexes A,B). We consider patient data in time as correlated feelings. The correlation is accorded with patient-related aggregated features and dependent variables, which are both constant for each patient. Symptoms before months (at most 1 year) of each time questionnaire are named past symptoms.

4.5 Prediction approach using clustering and classification

We present in the following two approaches for identifying the correlation between patient QoL-related data. The clustering approach is tested followed by the classification approach. For both approaches, we fusion HCB and IPOL datasets using the HCB attributes labels.

4.5.1 Clustering

We evaluated seven clustering algorithms—Birch, K-means, Gaussian Mixture, Robust K-means, Agglomerative Clustering, K-medoids, and Minibatch K-means—using a combined dataset from two hospitals. The methodology employed in this study builds upon the approach introduced by (Agoun et al., 2023), specifically designed to optimize clustering for Quality of Life (QOL) data. In unsupervised learning, one of the central challenges is determining the optimal number of clusters due to the absence of predefined labels or ground truths. This methodology systematically addresses this issue by integrating dimensionality reduction and clustering performance evaluation.

The approach involves two primary stages. First, dimensionality reduction is performed using UMAP (Uniform Manifold Approximation and Projection), a non-linear technique renowned for preserving both local and global structures of high-dimensional data when embedding it into lower-dimensional spaces. This step is critical in unsupervised learning, as it mitigates the curse of dimensionality and improves the efficiency of clustering algorithms.

In this study, the combined dataset comprised 210 *time patients* and 51 attributes. Using the proposed methodology, we explored dimensionality reductions to 2, 3, 6, 8, 10, 12, and 14 dimensions. All seven clustering algorithms were applied, and their performances were evaluated across these dimensional settings. The results are presented in Table 4.

The results indicated that reducing the data to six dimensions produced the lowest variance in the number of clusters, with an average of six clusters. This finding suggests that the inherent structure of the data is most consistently captured at six dimensions, providing a reliable basis for identifying the optimal number of clusters.

Consequently, we assume that the optimal number of clusters for our data is six. Once this optimal number was established, we analyzed the characteristics of the clusters generated by each algorithm. For this purpose, we employed the feature extraction algorithm proposed by (Khoie et al., 2017), which identifies the most representative variables for each cluster, thereby enhancing the interpretability of the clustering results. The extracted features for each cluster provided deeper insights into the underlying patterns in the dataset, enabling more informed interpretations and applications of the findings.

4.5.2 Clustering Results

Table 5 presents Birch and k-means algorithms results. Each attribute is characterized by a high (+) or low value (-). Please refer to the annexes A and B for attribute values. Additional attributes are specified by an algorithm in a cluster, we mention the algorithm in brackets. Bold attributes correspond to aggregated features.

The two algorithms identify 6 clusters as follows (see Table 5): *(C1)* Consuming cannabis without children melanoma patients with past reflux medications, *(C2)* Lymphoma patients with high past psychological and fatigue issues, *(C3)* Melanoma patients with children and with low past pain and reflux medications, *(C4)* Melanoma and Lymphoma patients with nausea, pain after immunotherapy and low past fatigue and psychological issues, *(C5)* Lymphoma young patients with high past fatigue, *(C6)* Consuming cannabis without children melanoma patients with past fatigue and psychological medications.

We detail the results of clusters. C1 and C6 have common attributes (see Table 5). They both describe melanoma patients without children, who consume high amount of cannabis, and feel a lot of anxiety. During the whole survey, they are averagely content of their QoL, but they feel low happiness (cheerful). C1 defines more these patients as consumed highly past reflux medications. Additionally, Birch algorithm specifies that these patients had low past fatigue and received a high number of therapies (*i.e.* immunotherapies, other therapies). C6 specifies melanoma without children patients as administered high amounts of past fatigue and psychological medications (stress, anxiety). They received a high number of immunotherapies, and a low number of doses of immunotherapies.

C3 describes melanoma patients with children. They express happiness and have low anxiety (frightened feelings, worrying thoughts). They

Dimension	Variance	Average Nb Clusters
2D	0,57	6
3D	2,67	7
6D	0,24	6
8D	0,33	8
10D	4,50	8
12D	5,33	6
14D	4,48	6

Table 4: Variability and Connectivity Results by dimension

<table>
<tr><th>C1</th><th>C6</th><th>C2</th></tr>
<tr><td colspan="2">Hospital of inclusion +
Cancer Type +
No Children +
Cannabis Consumption +
Frightened Feelings -
Worrying Thoughts -
Cheerful -
QoL Contentment +</td><td rowspan="2">Hospital of inclusion -
Cancer Type -
Past sadness and depression +
Past stress and anxiety +
Past fatigue impact +
Past sleep disorder impact +
Past sadness and depression impact +
Past stress and anxiety impact +
Past pain impact + (K-means)
QoL Contentment -
Average number of immunotherapies -
Average days interval of immunotherapies -</td></tr>
<tr><td>Past fatigue - (Birch)
Past Reflux Medications +
Average Number of Therapies + (Birch)</td><td>Feeling tense -
Past fatigue medications +
Past sadness and depression medications + (K-means)
Past stress and anxiety medications +
Average number of immunotherapies +
Average days interval of immunotherapies +</td></tr>
<tr><th>C3</th><th>C4</th><th>C5</th></tr>
<tr><td>Hospital of inclusion +
Cancer Type +
Children -
Frightened Feelings +
Worrying Thoughts +
Cheerful +
Past pain medications -
Past reflux medications -</td><td>Nausea +
Trouble meeting family needs +
Pain +
Enjoying things +
Laugh +
Cheerful -
Past fatigue -
Past stress and anxiety -
Past fatigue impact -</td><td>Hospital of inclusion -
Birth Date +
Cancer Type -
Cheerful +
Past fatigue +
Past fatigue impact +
Disease Status +
Average number of immunotherapies -
Average days interval of immunotherapies -</td></tr>
</table>

Table 5: Common Clusters Results of QoL-patient data (**Birch and K-means** algorithms)

used few pain and reflux medications in the past. C2 cluster contains lymphoma patients with high past impacting psychological issues (sadness, depression, stress, anxiety), sleep disorders, and fatigue. They received a low number of immunotherapies and a high number of doses of immunotherapies. They express low contentment of their QoL. C4 concerns both lymphoma and melanoma patients who experienced high nausea and pain after immunotherapies. They had low past fatigue and psychological issues (stress, anxiety). They express low happiness (laughing, enjoying things, cheerful) and have trouble meeting their family needs. C5 describes Lymphoma young patients who express happiness (cheerful). They received a low number of immunotherapies and high number of doses of immunotherapies. They averagely responded well to therapies (immunotherapies and other therapies). They experienced high past fatigue impacting their daily life.

We define the correlations of symptoms in clusters. The type of cancer and the hospital are correlated, as the collected data concern one type of cancer per hospital. Cheerfulness, worrying thoughts, and frightened feelings are highly correlated symptoms in clusters. We describe the determinants of QoL-related data. The aggregated attribute contentment of QoL appears in 3 clusters out of 6 (see Table 5). The aggregated number of therapies is expressed in 4 clusters. Aggregated Disease status appears in 1 cluster. 4 other unaggregated attributes are key attributes in the data as follows: Cheerfulness and past fatigue appear in 5 clusters out of 6. Anxiety and past anxiety attributes appear in 5 clusters.

We present in the following the results of clustering of the other 5 clustering algorithms (*i.e.* Gaussian-Mixture (GM), Robust k-means, Agglomerative clustering, K-Medoids, and Minibatch k-means). Results are presented in terms of difference with the clusters defined by Birch and k-means.

Table 6 presents additional attributes to the clusters defined by Birch and k-means. Some attributes substitute others in the six clusters of Birch and k-means, we mark them in the table 6 with *. Bold attributes correspond to aggregated features.

We resume additional findings of these 5 algorithms in the following. Consuming cannabis without children melanoma patients (C1 and C6) have low sleep disorder issues. C2 cluster patients are young and anxious. Their Adverse Events (AE) are related to administrated drugs and they averagely did not respond well to other therapies (chemo/radiotherapies). C3 cluster of melanoma patients living with children are expressing happiness. They have low anxiety and low nausea symptoms and are content with their QoL. A subgroup of them received a high number of therapies (Agglomerative clustering). A second subgroup has suffered from past impacting fatigue, sleep disorders, and psychological issues (Agglomerative clustering). However, the two subgroups of C3 generated by Agglomerative clustering are not opposite. C4 cluster describing Melanoma and Lymphoma patients with nausea, and pain after immunotherapy is more characterized by GM and Robust k-means algorithms (see Table 6). These patients averagely responded badly to immunotherapy and responded well to other therapies. Their AE are not related to drugs. K-Medoids and Mini-Batch algorithms describe Melanoma and Lymphoma patients without children with nausea and pain after immunotherapy as consumers of cannabis. They have low past sleep disorders.

4 algorithms out of 5 describe C5 Lymphoma young patients as practicing physical activity. K-Medoids specify also that C5 patients have a high level of education. They received more other therapies than immunotherapies and they responded badly to other therapies. C6 patients responded averagely badly to immunotherapies (GM and Robust k-means). Mini-Batch k-means specify that the C6 patients are old and responded badly to other therapies. One contradiction between algorithms in C6 is that patients did not highly suffer from past stress and anxiety with Mini-Batch, but they did take high amount of stress and anxiety medications with Birch and k-means.

Extra clusters are specified by the 5 algorithms (see Table 7). We describe the extra informative clusters referring to their similitude with Birch and k-means clusters using the same numeration of clusters in Table 5, with extra quotes.

(C4') Old-aged melanoma and lymphoma patients with nausea and lack of energy. They express low happiness (Cheerful, Enjoying things, see Table 7), and are anxious (Worrying thoughts). They responded on average well to immunotherapies. Their AE are related to administered drugs. They received high amounts of past pain, fatigue, and psychological (stress, anxiety) medications. A contradiction is that their past fatigue and psychological issues are expressed as low. This cluster is close to C4 (Birch and k-means, see Table 5). However, C4 expresses patients with nausea and pain, and C4' describes old patients with nausea and lack of energy.

K-Medoids and Mini-Batch algorithms specify an extra characteristic of QoL patients: 'health discussions with their partner'. Mini-Batch defines *(C3')* cluster as melanoma old patients living with children, having a lot of health discussions with their partner. They are content of their QoL. Their AE are related to administered drugs. They did not respond well to other therapies. *(C2')* resembles to C2 cluster adding high anxiety to past high anxiety and other psychological issues of C2 lymphoma patients. These patients responded well to immunotherapy in C2'. Mini-Batch algorithm describes another extra cluster *(C5')* Lymphoma patients living with children. They have a lot of health discussions with their partner.

To conclude the results, the 'AE-relationship with drug' aggregated feature was absent in Birch and k-means. This can be because our data contains few clinical and AE-related data. 'Disease Status' aggregated feature was more present in GM, Robust k-means, and Mini-Batch k-means. 'QoL contentment' aggregated feature was specified by all the algorithms.

4.5.3 Classification

We use supervised learning *classification algorithms* (using scikitlearn) as a prediction model. Characteristic normalization is performed using the Standard Scaler technique. Missing values are imputed using K nearest neighbors imputer. K is set to 3. The number of features is 49. The following six classification algorithms are chosen for data training: Random Forest, SVM, Logistic Regression, GBM, Gaussian Process, and MLP Neural Network. A function for selecting the N salient variables using the Random Forest algorithm is implemented. N is set to 10. We evaluate the models with the corresponding salient attributes. For each

	GM & Robust k-means	Agglomerative	K-Medoids	Mini-Batch
C1	X	Feeling tense -		Past sleep disorders - Past fatigue medications + Past stress and anxiety medications +
C2	Birth date +		Worrying thoughts - **AE-relationship drugs +** DS other therapies +	X
C3	Nausea - Enjoying things - **QoL Contentment +**	* Living with children + * Feeling tense + *** QoL Contentment +** * Average number of therapies + Nausea - Trouble meeting family needs - Enjoying things - Past sleep disorders + Past fatigue impact + Past sleep disorder impact + Past stress and anxiety impact +	X	
C4	**AE-relationship drugs - DS** - DS immuno + DS other therapies -	* Lack of energy +	* Children + * Cannabis Consumption + * Past sleep disorders -	
C5	Physical Activity +		* Physical Activity + * Level of education + * Nausea - * DS other therapies + * Average number of therapies +	X
C6	Past reflux medications + * Past fatigue - *** DS immuno +**	X		* Birth date - * Living with children + * Past sleep disorders - * Past stress and anxiety - * DS other therapies +

Table 6: Additional attributes by the other clustering algorithms

dependent variable, we select the best model.

4.6 Classification Results

In the following, we present the results of the six tested algorithms. We present the accuracy of each algorithm according to the dependent variable (see Table 8). The GBM model should be used among the six algorithms tested to predict the 3 dependent variables. Each dependent variable corresponds to the top ten features(see Table 9). Common features of the 3 dependent variables are the average number of medications, date of birth, and the frequency of immunotherapies.

Anxiety and *sadness-related attributes* are in the top features for QoL contentment, Adverse Events relationship with drug, and disease evolution.

4.7 Prediction Model

The pre-trained data of HCB and IPOL datasets, the associated metadata, and the new data consisting of 10 features are used to predict the 3 dependent variables. An example of input consisting of at least 10 attributes corresponding to the top 10 features. The input is specified in natural language as follows "I have worrying thoughts. I had stressful feelings before immunotherapy but not that much. I don't take anxiety medications. My cancer progressed with previous chemotherapies and responded partially with immunotherapies. Cancer is melanoma. I took about 2 immunotherapies and 3 chemotherapy treatments. I graduated with a master of science. I am 48 years old. I take daily about 4 medications".

The output is expressed in terms of three indicators:

- Low value of *QoL contentment indicator* correlated with a high value (superior to 2) of the *disease status* and a zero value of *AE relationship with drug* alerts about more focus on psychological support,
- Progression of disease (*disease status indicator*, value 2) correlated with a high value of *QoL contentment* and zero value of *AE relationship with drugs* alerts about a further

Gaussian-Mixture and Robust k-means	**K Medoids**	
C4'	**C3'**	**C3"**
Birth date - Lack of energy + Nausea + Enjoying things + Worrying thoughts - Cheerful - Past fatigue - Past sleep disorders - Past stress and anxiety - Past fatigue medications + Past stress and anxiety medications + Past pain medications + **AE-relationship drugs** + **DS Immuno** -	Hospital of inclusion + Cancer Type + Health discussions with partner - Nausea - Trouble meeting family needs - Enjoying things - Laugh - Cheerful + **QoL Contentment +**	Hospital of inclusion + Cancer Type + Living with children + **QoL Contentment +** Average number of therapies +
Mini-Batch		
C2'	**C3'**	**C2"**
Hospital of inclusion - Cancer Type - Frightened Feeling - Worrying thoughts - Past sadness and depression impact + **AE-relationship drugs +** **QoL Contentment -** **DS immuno -** **DS other therapies +** Average number of immunotherapies - Average days interval of immunotherapies -	Hospital of inclusion + Birth date - Cancer Type + Health discussions with partner - Living with children + Trouble meeting family needs - Bothering side effects - **AE-relationship drugs +** **QoL Contentment +** **DS other therapies +**	Hospital of inclusion - Cancer Type - Health discussions with partner - Detailed health discussions with partner - Living with children + Feeling tense + **DS other therapies +** Average number of immunotherapies - Average days interval of immunotherapies -

Table 7: Different Clusters Results of QoL-patient data by the other algorithms

	QoL contentment	Drugs	Disease
Random Forest	0.7363 (+/- 0.0802)	0.8810 (+/- 0.0410)	**0.8952** (+/- 0.0375)
SVM	0.6313 (+/- 0.0471)	0.8190 (+/- 0.0683)	0.7667 (+/- 0.0135)
Logistic Regression	0.5934 (+/- 0.0159)	0.8095 (+/- 0.0587)	0.8000 (+/- 0.0233)
GBM	**0.8179** (+/- 0.0487)	**0.8857 (+/- 0.0309)**	0.8952 (+/- 0.0178)
Gaussian Process	0.5689 (+/- 0.0652)	0.8286 (+/- 0.0649)	0.7190 (+/- 0.0067)
MLP Neural Network	0.5932 (+/- 0.0252)	0.8571 (+/- 0.0309)	0.8714 (+/- 0.0202)

Table 8: Classification models accuracy

inspection of the actual therapy,

- Positive *AE relationship with drugs* indicates an inspection of the administrated medications.

5 Conclusion

We were interested in this work in predicting QoL determinants of cancer patients that received immunotherapy treatment. For this, we presented a state-of-the art of existing approaches for determining QoL key attributes and their correlation with other key health symptoms. The literature studies are not consolidated for summarized consensual or cumulative QoL prediction models of cancer patients (Kaplan and Hays, 2022). For this, we proposed to consolidate existing works by using both clustering and classification models. To ameliorate explainability of these models, we tried to focus more on patient-reported feelings rather than a mix of patient-clinicians data. Also, we used raw QoL attributes of the FACT-G assessment tool instead of using QoL composite scales. The used datasets are based on correlated QoL data and some clinical data for a period of 18-months post-immunotherapy. We mapped two european hospital datasets (HCB and IPOL) to predict 3 key indicators (QoL contentment, disease status and Adverse Events relationship with drugs). A patient has several QoL-related data through time. These data were correlated using his temporal aggregated key features. Our identified identified features using clustering models helped identify general QoL-determinants and their correlation with other clinical data. Also, using classification models by setting 3 key indicators help towards a more con-

QoL Contentment	**AE-relationship Drugs**	**Disease Status (DS)**
Average number of medications	Birth date	DS Immuno
Birth date	DS other therapies	Birth date
Average days interval of immunotherapies		
Bothering side effects	Average number of medications	
Past sadness and depression impact	DS Immuno	Average number of all therapies
Trouble meeting family needs	Past stress/anxiety medications	DS other therapies
Enjoying things	Worrying thoughts	Cancer type
Laugh	Level of education	Hospital of inclusion
Nausea	Enjoying things	Past stress and anxiety
DS other therapies	Average number of therapies	Level of education Level

Table 9: Dependent Variables Top-10 Features

solidated prediction model based mainly on QoL patient data.

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A IPOL and HCB datasets common Raw Attributes Dictionary

B IPOL and HCB datasets common Aggregated Clinical Attributes Dictionary

Attribute	Values
Socio-demographic and clinical data (constant)	
1. Questionnaire Time	0, Baseline | 1, 3 months | 2, 6 months | 3, 9 months | 4, 12 months | 5, 18 months | 6, 24 months | 7, Progression
2. Hospital of inclusion	1, IPOL | 2, HCB
3. Date of birth	Year
4. Sex	1, Male | 2, Female
5. Cancer Type	1, Lymphoma | 2, Melanoma
6. Children	1, Yes | 2, No
7. Level of education	1, No | 2, Primary | 3, Lower secondary | 4, Higher secondary | 5, Post-secondary non-tertiary | 6, Short-cycle tertiary | 7, Bachelor | 8, Master | 9, Doctoral
Actual QoL-related data (variable at each questionnaire time)	
8. Living with children	1, Yes | 2, No
9. Health discussions with partner	1, Every time | 2, Often | 3, Occasionally | 4, Rarely | 5, Never
10. Detailed Health discussions with partner	
11. Cannabis Consumption	1, Never | 2, Less than once a month | 3, Once a month | 4, Once a week | 5, Every day or nearly
12. Physical Activity	1, Yes, every day | 2, Yes, not every day | 3, No
13. Daily Care of another sick person	1, Yes | 2, No
14. Lack of energy	1, Not at all | 2, A little bit | 3, Somewhat | 4, Quite a bit | 5, Very much
15. Nausea	
16. Trouble meeting family needs	
17. Pain	
18. Bothering Side Effects	
19. Feeling ill	
20. Feeling Tense	1, Most of the time | 2, A lot of the time | 3, From time to time, occasionally | 4, Not at all
21. Enjoying Things	1, Definitely as much | 2, Not quite a bit | 3, Only a little | 4, Hardly at all
22. Frightened Feeling	1, Very definitely and quite badly | 2, Yes, but not too badly | 3, A little, but it doesn't worry me | 4, Not at all
23. Laugh	1, As much as i always could | 2, Not quite so much now | 3, Definitely not so much now | 4, Not at all
24. Worrying thoughts	1, A great deal of the time | 2, A lot of the time | 3, From time to time, but not too often | 4, Only occasionally
25. Cheerful	1, Not at all | 2, Not often | 3, Sometimes | 4, Most of the time
Past QoL-related data (variable at each questionnaire time)	
26. Past Fatigue	1, Never | 2, Rarely (less than twice a month) | 3, Occasionally (2-4 times a month) | 4, Often (more than once a week) | 5, Every day
27. Past Sleep Disorders	
28. Past Sadness and Depression	
29. Past Stress and Anxiety	
30. Past Pain	
31. Past Reflux	
32. Past Fatigue impact	1, Not at all | 2, Rarely | 3, To some extent | 4, Frequently | 5, Greatly
33. Past Sleep Disorders Impact	
34. Past Sadness and Depression Impact	
35. Past Stress and Anxiety Impact	
36. Past Pain impact	
37. Past Reflux impact	
38. Past medications (excluding cancer treatments)	1, None | 2, 1-5 medications a day | 3, 5-10 medications a day | 4, 10 medications or more a day
39. Past Fatigue Medications 40. Past Sleep Disorders Medications 41. Past Sadness and Depression Medications 42. Past Stress and Anxiety Medications 43. Past Pain Medications 44. Past Reflux Medications	1, Never | 2, Rarely (less than twice a month) | 3, Occasionally (2-4 times a month) | 4, Often (more than once a week) | 5, Every day

Attribute	Values
Aggregated Clinical Attributes per patient for all time questionnaires (constant)	
45. AE-relationship with drug (mean)	AE-relationship with drug:'Yes':1, 'No': 2, 'Possibly': 3 (raw data) $$\frac{\text{Number of 'Yes'} + \text{Number of 'Possibly'}}{\text{Number of 'Yes', 'No', 'Possibly' per patient}} \quad (1)$$ Output 0:AE-Unrelated to medications, 1: AE-related to medications
46. Disease Status (DS)	DS for each Questionnaire Time: 1, Stable disease | 2, Progressive disease | 3, Partial response | 4, Complete response $$\text{Average(Disease Status at each Time)} \quad (2)$$
47. DS Immunotherapies	DS for each immunotherapy: 1, Complete response | 2, Partial response | 3, Stable disease | 4, Progressive disease $$\frac{\text{Disease Status for each immunotherapy}}{\text{Number of Disease Status for all Immunotherapies of the patient}} \quad (3)$$
48. Disease Status Other therapies	Disease Status for each chemo/radiotherapy: 1, Complete response | 2, Partial response | 3, Stable disease | 4, Progressive disease $$\frac{\text{Disease Status for each chemo/radiotherapy}}{\text{Number of Disease Status for all chemoradiotherapies of the patient}} \quad (4)$$
49. Average Number of Medications	Medication type: 1, Inmunosuppresive | 2, Medication to treat autoimmune endocrinopaties | 3, None of the others $$\frac{\text{Number of Medications per patient}}{\text{Average Number of Medications per Hospital}} \quad (5)$$
50. Average Number of Therapies	Therapy Type: Textual $$\frac{\text{Number of Therapies per patient}}{\text{Average Number of Therapies per Hospital}} \quad (6)$$
51. Average Number of Immunotherapies	Therapy Type: Textual $$\frac{\text{Number of Immunotherapies per patient}}{\text{Number of Therapies per patient}} \quad (7)$$
52. Average Days Interval of Immunotherapies	Therapy Frequency: Interval of days between medications (Textual) $$\frac{\text{Total Frequencies of immunotherapies per patient}}{\text{Number of Immunotherapies per patient}} \quad (8)$$
Aggregated QoL Attribute per patient for all time questionnaires (constant)	
53. QoL Contentment	I am content with my QoL :1, Not at all | 2, A little bit | 3, Somewhat | 4, Quite a bit | 5, Very much $$\text{Average(I am content with my QoL for each Time)} \quad (9)$$